

Chapter 2

Literature Review

Literature Review: RAG & Agentic AI in Education

The development of the **AI-Powered Educational Assistant** is based on recent advancements in artificial intelligence, educational technology, intelligent retrieval systems, and autonomous reasoning workflows. Modern learning environments increasingly rely on AI to improve student engagement, personalize education, and automate academic support tasks such as summarization, assessment generation, and question answering.

Traditional e-learning platforms usually provide static content with limited interaction. In contrast, AI-powered educational systems can create dynamic and adaptive experiences where students interact directly with intelligent assistants capable of understanding context and responding to individual needs.

This project specifically builds upon three major research directions: **Retrieval-Augmented Generation (RAG), Agentic AI systems, and AI evaluation frameworks.**

Retrieval-Augmented Generation (RAG) in Education

Traditional Large Language Models (LLMs) may generate fluent responses, but they often suffer from factual inaccuracies or hallucinations when answering domain-specific academic questions. This creates risk in educational applications where correctness is essential.

Retrieval-Augmented Generation (RAG) solves this challenge by combining language models with an external retrieval system. Before generating an answer, the system searches trusted sources such as uploaded PDFs, lecture notes, or course materials, then uses the retrieved context to produce grounded responses.

For educational systems, RAG provides several benefits:

- Accurate answers based on student documents
- Context-aware explanations from course materials

- Reduced hallucinations
- Better trustworthiness in academic settings
- Ability to answer from custom user content rather than only pretrained knowledge

This makes RAG a highly suitable architecture for AI study assistants.

Agentic AI and Multi-Agent Workflows

Modern AI systems are moving beyond simple one-step question answering toward **Agentic AI**, where the model performs planning, reasoning, tool usage, and multi-step execution.

This project uses a **Multi-Agent Workflow** composed of several specialized components:

Agent	Role
Planner	Understands the user request and creates execution steps
Retriever	Searches uploaded materials and relevant context
Executor	Generates summaries, quizzes, explanations, or answers
Verifier	Checks quality, consistency, and hallucination risk

This structure mimics human problem-solving and allows the system to handle complex tasks such as:

- Generating adaptive quizzes
- Producing structured summaries
- Creating personalized study plans
- Handling multi-intent requests in one interaction
- Verifying outputs before returning them to the user

Agent-based systems improve flexibility, scalability, and reasoning quality compared to standard chatbot models.

AI Evaluation and Hallucination Mitigation

In educational applications, AI outputs must be measured carefully to ensure safety, quality, and reliability.

Recent frameworks such as **RAGAS** are widely used to evaluate RAG pipelines using metrics such as:

Metric	Purpose
Faithfulness	Measures whether answers are supported by retrieved context
Context Precision	Measures relevance of retrieved content
Context Recall	Measures completeness of retrieved content
Answer Relevancy	Measures how well the response answers the query

To maintain educational reliability, the system should also include:

- Guardrails for unsafe or invalid outputs
- Confidence scoring
- Fallback responses when evidence is weak
- Verification loops before final generation
- Prompt testing and regression evaluation

These controls are critical to making AI dependable in academic environments.

Suggested Improvements – Areas Where the Project Can Be Enhanced

Although the current scope is strong, several future improvements can transform the project into a production-level platform.

Area	Suggested Improvement
Enterprise AI	Integrate enterprise-grade providers such as OpenAI Enterprise or Claude
Privacy & Compliance	Add institutional data governance and privacy controls

Multimodal Learning	Support images, handwritten notes, OCR, audio, and video transcripts
LMS Integration	Connect with Moodle, Blackboard, or university portals
Collaboration	Real-time shared study rooms and group quizzes
Predictive Analytics	Detect struggling students early using learning data
Gamification	Add badges, streaks, achievements, and rewards
Mobile Apps	Native iOS and Android applications
Accessibility	Voice assistant, multilingual support, dyslexia-friendly modes
Advanced Personalization	Deeper adaptive learning recommendations

Final Grading Criteria

The final project grade may be distributed across four major categories.

Evaluation Category	Weight	Description
Documentation (Report & Research)	20%	Literature review, diagrams, methodology, academic writing
Implementation (Software & AI Core)	40%	Frontend, backend, AI modules, deployment, scalability
Testing & Evaluation	20%	Functional testing, AI metrics, system performance
Presentation & Defense	20%	Live demo, technical explanation, teamwork, communication

Total = 100%

Overall Academic Value

This graduation project is considered highly valuable because it combines:

- Artificial Intelligence

- Full-stack Software Engineering
- Educational Technology
- Research Evaluation Methods
- Real-world User Impact

With strong execution, testing, and documentation, the project has excellent potential for a high final grade and future real-world expansion.